

System Development Approach – Azure + Claude AI (Brief)

1. Cloud Infrastructure – Azure

- **Hosting** on Azure App Service, Azure Functions, and Azure PostgreSQL.
 - **Device sync** via Azure IoT Hub (biometric locks, smart kiosks).
 - **Blob Storage** for media (logs, receipts, photos).
 - **Azure AD B2C** for secure user authentication and role-based access.
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2. Rapid Prototyping with Claude AI

- Use **Claude AI** (Anthropic) for:
 - Fast generation of UI wireframes, API scaffolding, and system documentation.
 - Logic prototyping for wallet transactions, access rules, and business flows.
 - Auto-generating test cases and data models for backend APIs.
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3. Modular Components

- **Biometric Access:** OpenBR + Azure IoT Edge for device control.
 - **Wallet System:** Custom service integrated with mobile money APIs (MTN, Airtel).
 - **Attendance & Time Logs:** Open Time Clock, hosted and connected to Azure.
 - **Accounting:** ERPNext on Azure VM or container.
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4. Agile Development Pipeline

- Code hosted in **GitHub**, CI/CD with **Azure DevOps**.
 - Deploy staging & production environments on Azure.
 - Use Claude AI to draft user stories, code snippets, and automation scripts.
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5. Frontend & Notifications

- **React.js** (Admin Portal), **Flutter** (Mobile App).
 - Push alerts via **Azure Notification Hubs**.
 - SMS/USSD via Africa's Talking or Twilio.
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System Diagram



